PyHelmholtz User Manual

version 0.2.0

PyHelmholtz is a Python package for solving the two-dimensional, time-harmonic Helmholtz equation and comparing absorbing boundary methods (ABMs) in a common finite-difference framework. Its main components are the computational domain, source, absorbing boundary treatment, finite-difference scheme, and solver.

Features

- Frequency-domain wave propagation modeling.
- Support for standard sparse direct solvers via SciPy.
- Optional high-performance nested dissection algorithms via SuiteSparse.
- Optional parallel direct solver acceleration via MUMPS.

Installation

PyHelmholtz provides three installation options depending on your performance needs and platform. Python 3.10 or later is required.

1. Standard Installation (Recommended)

If you only need standard solvers using SciPy's `spsolve()`, you can install PyHelmholtz instantly with no external C-library or compiler requirements:

```
pip install pyhelmholtz
```

2. With Nested Dissection Support (Optional)

To enable advanced nested dissection algorithms, the package requires `scikit-sparse`. Because this depends on underlying SuiteSparse C-libraries, it is highly recommended to install the dependencies via Conda first to avoid compilation issues (especially on Windows):

```
# Install the C-libraries and pre-compiled wrapper via Conda
conda install -c conda-forge scikit-sparse

# Install PyHelmholtz with the optional suitesparse feature flag
pip install pyhelmholtz[suitesparse]
```

3. With MUMPS Acceleration (Optional, Linux only)

For large-scale problems requiring parallel direct solvers, PyHelmholtz interfaces with MUMPS via PyMUMPS. Because this relies on compiled Fortran, MPI, and BLAS/LAPACK binaries, you must install the system MUMPS libraries first before installing the Python package:

```
# Install system MUMPS and MPI development libraries (Ubuntu/Debian)
sudo apt-get update
sudo apt-get install libmumps-dev

# Install PyHelmholtz with the optional mumps feature flag
pip install pyhelmholtz[mumps]
```

Quick Start

Example 1: Default configuration

Here is a quick example of how to use PyHelmholtz to solve the Helmholtz equation using the default parameters. The default parameters are summarized in Table 1.

Parameters	Values
physical domain Ω_i	[-1, 1] × [-1, 1] m^2
velocity $v(x, y)$	299792458 m/s (homogeneous)
source type	point source
source location	(0, 0) m
source frequency f	2 GHz
source strength S	1
grid spacing h_x and h_y	0.01 m ($h_x = h_y$)

Numerical configurations	Settings
FD accuracy order	2
absorbing boundary method	second-order Engquist-Majda (EM2) (one-cell boundary layer)
total grid points, $N_x \times N_y$	203 × 203
linear solver	SuperLU (`scipy.sparse.linalg.spsolve()`)

Table 1: Default parameters and numerical configurations of the `Helmholtz` class.

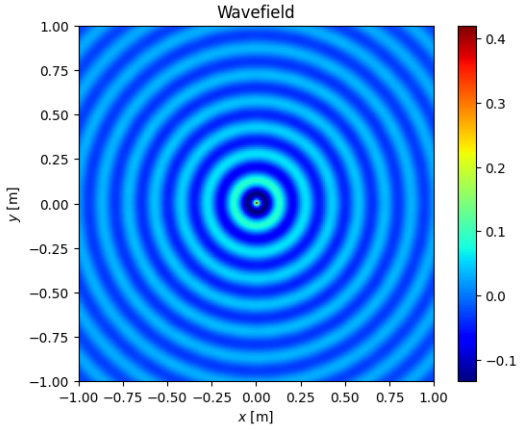


Figure 1: The computed wavefield.

This example uses the package defaults. These defaults define a homogeneous medium with a point source at the origin and the EM2 absorbing boundary treatment. The computed wavefield, as shown in Fig. **??**, is stored in the object `ho` and can be accessed using `ho.u`.

```
import pyhelmholtz as ph

ho = ph.Helmholtz()
ho.solve()
ho.viz()
```

Example 2: Custom parameters

This example defines the domain, source, absorbing boundary method, and finite-difference order separately, then passes them to the main solver.

The physical domain is defined by the rectangle $[-500, 500] \times [-200, 300]\,\mathrm{m}^2$ discretized with a grid spacing of $h = 5\,\mathrm{m}$ in both directions. The background wave speed is uniform at 1000 m/s. An object of size $[-200, 50] \times [-100, 100]\,\mathrm{m}^2$ is included with a wave speed of 800 m/s. The source has a frequency of 10 Hz and is positioned at $(200, 200)\,\mathrm{m}$. The number of PML cells is 15, and a fourth-order finite-difference scheme is used for spatial discretization. Ensure that the source lies within the physical domain and that the chosen grid adequately resolves the wavelength ($\lambda_{\min}/h = 16$).

```
import pyhelmholtz as ph
import matplotlib.pyplot as plt

domain = ph.Domain(limits=(-500, 500, -200, 300), h=5, v=1000)
domain.add_rectangle(bounding_box=(-200, 50, -100, 100), v=800)

source = ph.PointSource(freq=10, xs=200, ys=200)
abm = ph.PML(n=15)
fd = ph.FD(order=4)

ho = ph.Helmholtz(domain=domain, abm=abm, source=source, fd=fd)

ho.solve()

plt.subplots(1, 2, figsize=(9, 3))
plt.subplot(121); ho.viz(data="velocity")
plt.subplot(122); ho.viz(data="solution")
plt.tight_layout()
plt.show()
```

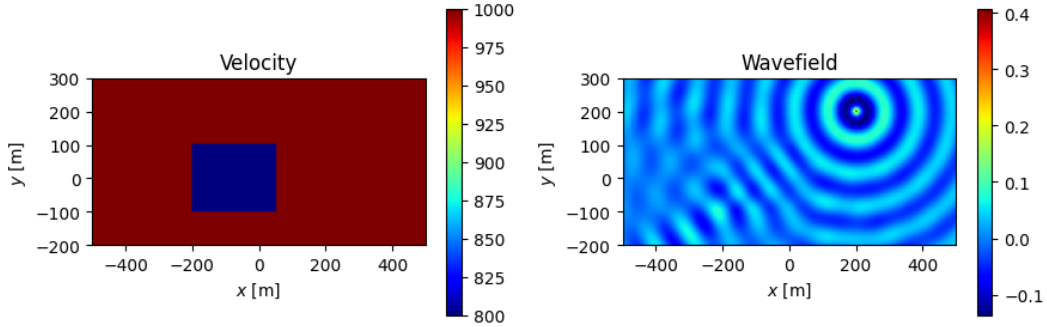


Figure 2: The velocity model and the computed wavefield.

API Reference

1. Domain

The `Domain` class defines the 2D physical domain, its spatial grid, and background wave-speed model. It also provides utility methods for embedding geometric velocity anomalies into the grid.

```
ph.Domain(limits=(-1,1,-1,1), h=0.01, v=299792458, positive_downward=False)
```

Parameter	Data type	Description	Default
`limits`	tuple of float	Physical domain $[x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}]$ [m]	`(-1, 1, -1, 1)`
`h`	float or array-like	Grid spacing in the x- and y-directions [m]	`0.01`
`v`	float or array-like	Background wave speed [m/s]; scalar (homogeneous) or spatially varying (inhomogeneous)	`299792458` (light speed)
`positive_downward`	bool	If `True`, the y-axis points downward	`False`

The `Domain` class provides two methods for defining objects in the physical domain: `Domain.add_circle()` and `Domain.add_rectangle()`.

The method `Domain.add_circle()` assigns a specified value to grid points inside or on the boundary of a circle.

```
domain = ph.Domain()
domain.add_circle(center, radius, v)
```

Parameter	Data type	Description	Default	Example
`center`	tuple of float	Center (x_c, y_c) [m]	–	`(-1.5, 1.5)`
`radius`	float	Radius [m]	–	`3.2`
`v`	float	Wave speed (homogeneous) [m/s]	–	`2e8`

The method `Domain.add_rectangle()` assigns a specified value to grid points within a rectangular region defined by $[x^b_{\min}, x^b_{\max}] \times [y^b_{\min}, y^b_{\max}]$, including the endpoints.

```
domain.add_rectangle(bounding_box, v)
```

Parameter	Data type	Description	Default	Example
`bounding_box`	tuple of float	Bounding box limits $[x^b_{\min}, x^b_{\max}] \times [y^b_{\min}, y^b_{\max}]$ [m]	–	`(-0.3, 0.3, -0.8, -0.2)`
`v`	float	Wave speed (homogeneous) [m/s]	–	`25e8`

2. Source

The `Source` class defines source models for the 2D frequency-domain Helmholtz equation. `Source` is the abstract base class, and `PointSource` and `PlaneWave` are concrete implementations. These two classes construct the source vector $\mathbf{b}$ of the linear system $\mathbf{Ax} = \mathbf{b}$.

2.1 Point source

`PointSource` defines one or more point sources, $S_j\delta(x - x_{s,j}, y - y_{s,j})$, at coordinates $(x_{s,j}, y_{s,j})$ for the specified frequency `freq`, where S_j denotes the strength of the j-th source. The source locations and strengths determine the entries of the source vector $\mathbf{b}$.

```
ph.PointSource(freq=2e9, xs=0.0, ys=0.0, strength=1.0)
```

Parameter	Data type	Description	Default
`freq`	float	Source frequency [Hz]	`2e9`
`xs`	float or array-like	x-coordinates of the source(s) [m]	`0.0`
`ys`	float or array-like	y-coordinates of the source(s) [m]	`0.0`
`strength`	float or array-like	Source strength(s) [a.u.]	`1.0`

2.2 Plane wave

`PlaneWave` defines an incident plane wave propagating at an angle `theta` (in degrees) relative to the positive x-axis. The incident plane wave is expressed as

$$u^{\text{inc}}(x, y) = Se^{ik_x(x-x_{\text{zp}})}e^{ik_y(y-y_{\text{zp}})},$$

where S is the incident wave amplitude, k_0 is the background wavenumber, and $k_x = k_0 \cos\theta$ and $k_y = k_0 \sin\theta$ are the x- and y-components of the background wavenumber vector, respectively. The point $(x_{\text{zp}}, y_{\text{zp}})$ specifies the reference location for the wave phase.

```
ph.PlaneWave(freq=2e9, v0=299792458, strength=1.0, theta=0.0,
             x_zerophase=0.0, y_zerophase=0.0)
```

Parameter	Data type	Description	Default
`freq`	float	Incident-wave frequency [Hz]	`2e9`
`v0`	float	Background wave speed [m/s]	`299792458` (light speed)
`strength`	float	Incident-wave amplitude [a.u.]	`1.0`
`theta`	float	Propagation angle relative to the positive x-axis [degrees]	0.0
`x_zerophase`	float	x-coordinate of the phase reference point [m]	`0.0`
`y_zerophase`	float	y-coordinate of the phase reference point [m]	`0.0`

3. Absorbing Boundary Method (ABM)

The `ABM` class defines the interface for assembling a system matrix $\mathbf{A}$ with a selected boundary treatment. The implementations comprise `RenLiu` and `PML`.

3.1 Ren–Liu hybrid method

The `RenLiu` class supports the Engquist–Majda and Higdon absorbing boundary conditions, as well as Ren–Liu hybrid formulations.

```
# Higdon absorbing boundary method (n=1)
ph.RenLiu(abm="H2")

# Ren-Liu hybrid method with EM1
# Absorbing thickness: 15 cells
ph.RenLiu(abm="RLEM1", n=15)
```

Parameter	Data type	Description	Default
`abm`	string	Absorbing boundary method	`"EM2"`
`n`	integer	Number of absorbing layers	`1`
`damping_profile`	callable	Weighting profile for the Ren–Liu method (`n` ≥ 2)	`lambda i,n: (i/n)**2`

The supported ABM names for the `RenLiu` class and their allowed layer thicknesses `n` are listed in the table below.

`abm`	Absorbing boundary method	Allowed `n`
`"EM1"`	First-order Engquist–Majda	1
`"EM2"`	Second-order Engquist–Majda (default)	1
`"H2"`	Second-order Higdon ($\phi = 0°, 45°$)	1
`"RLEM1"`	Ren–Liu hybrid with first-order Engquist–Majda	≥ 2
`"RLEM2"`	Ren–Liu hybrid with second-order Engquist–Majda	≥ 2
`"RLH2"`	Ren–Liu hybrid with second-order Higdon	≥ 2

The parameter `damping_profile` specifies how the weighting coefficient varies across the absorbing layers. It accepts a callable that takes the layer index `i` and the total number of layers `n` as inputs. The default profile is quadratic.

3.2 Perfectly Matched Layer (PML)

The `PML` class supports the unsplit-field PML and PML–ABC hybrids.

```
# PML with second-order Engquist--Majda outer boundary
# PML thickness: 15 cells
# Target reflection coefficient: R0 = 1e-6
ph.PML(abm="PMLEM2", n=15, R0=1e-6)
```

Parameter	Data type	Description	Default
`abm`	string	Absorbing boundary method	`"PML0"`
`n`	integer	Number of PML layers (`n` ≥ 2)	`5`
`damping_profile`	callable	Damping profile for the PML method	`lambda i, n: 1/(1+1j*(sigma_max/omega)*(i/(n-1))**m)`
`R0`	float	Target normal-incidence reflection coefficient R_0	`1e-5`

The supported ABM names for the `PML` class and their allowed layer thicknesses `n` are listed in the table below.

`abm`	Absorbing boundary method	Allowed `n`
`"PML0"`	PML with Dirichlet termination (default)	≥ 2
`"PMLEM1"`	PML with first-order Engquist–Majda outer boundary	≥ 2
`"PMLEM2"`	PML with second-order Engquist–Majda outer boundary	≥ 2
`"PMLH2"`	PML with second-order Higdon outer boundary	≥ 2

The parameter `damping_profile` for the `PML` class defines how the complex stretching factor varies across the PML region. It accepts a callable that takes the layer index `i` and the total number of PML layers `n` as inputs.

For the x-direction, the complex stretching factor is defined as

$$s_x = 1 + \frac{\mathrm{i}\sigma_{\max}}{\omega}\left(\frac{x}{L}\right)^m,$$

where $\sigma_{\max}$ is the maximum damping coefficient, ω is the angular frequency, L is the PML thickness, and m controls the damping profile. The default profile is quadratic, with $m = 2$.

The target normal-incidence reflection coefficient R_0 and the PML thickness L determine the maximum damping coefficient [1]:

$$\sigma_{\max} = -\frac{(m+1)v\ln R_0}{2L},$$

where v is the wave speed and $L = (n-1)h$ is the physical PML thickness, excluding the outer boundary grid point.

4. Finite Difference Order

The `FD` class specifies the finite-difference order used to approximate spatial derivatives on the grid. Higher-order schemes can improve accuracy for sufficiently smooth, well-resolved fields, but require wider stencils and more neighboring grid points.

```
ph.FD(order=4)
```

Parameter	Data type	Description	Default
`order`	integer	finite-difference accuracy order (2 or 4)	`2`

5. Core Helmholtz Solver

The `Helmholtz` class binds together domain, source, absorbing boundary treatment, and finite-difference scheme. It manages matrix assembly, linear system solving, data visualization, and analytic verification.

```
ph.Helmholtz(domain=ph.Domain(), source=ph.PointSource(), abm=ph.ReLiu(), fd=ph
    .FD())
```

Parameter	Type	Description
`domain`	`Domain`	Defines the physical domain and wave-speed model
`source`	`PointSource`, `PlaneWave`	Defines the excitation source
`abm`	`RenLiu`, `PML`	Defines the absorbing boundary method
`fd`	`FD`	Defines the finite-difference discretization

The `Helmholtz` class provides four main methods: `build()`, `solve()`, `viz()` and `error_norm_point_source()`.

5.1 `build`

The method `build` assembles the discrete Helmholtz system matrix $\mathbf{A}$ and source vector $\mathbf{b}$ and returns them as a pair.

```
ho = ph.Helmholtz()

# A: Complex sparse system matrix (CSR format), shape (m, m)
# b: Complex source vector, shape (m,)
A, b = ho.build()
```

5.2 `solve`

The method `solver` builds and solves the linear system $\mathbf{Ax} = \mathbf{b}$ using the selected numerical solver. The computed wavefield is stored in the object's `u` attribute.

```
ho = ph.Helmholtz()
ho.solve() # use the default solver (spsolve)
ho.solve(solver="mumps") # use PyMUMPS, if installed
```

Parameter	Type	Description	Default
`solver`	string	Numerical solver	`"spsolve"`

The supported values of `solver` are listed in the table below.

`solver`	Description	Type of solver
`"spsolve"`	SciPy sparse direct solver	Sparse direct solver
`"mumps"`	Parallel sparse solver via PyMUMPS	Parallel sparse direct solver
`"nested_dissection"`	Nested-dissection reordering via CHOLMOD	Sparse direct solver
`"gmres"`	GMRES with shifted ILU preconditioning	Iterative solver

If the requested optional solver cannot be used, the implementation falls back to `spsolve()`.

5.3 `viz`

The method `viz` displays a 2D plot of the computed wavefield or velocity model. It can show the real part, imaginary part, or magnitude of the selected data.

```
ho = ph.Helmholtz()
ho.viz(data="solution", mode="real", unit="m",
       cmap="jet", vlim=None, title="",
       xlabel=True, ylabel=True)
```

The parameters accepted by `viz()` are summarized below.

Parameter	Type	Description	Available options	Default
`data`	string or array	Data to visualize	`"solution"`, `"velocity"`, `"incident"`, `"total"`, NumPy array	`"solution"`
`mode`	string	Quantity to display	Real part (`"real"`), Imaginary part (`"imag"`), Magnitude (`"abs"`)	`"real"`
`unit`	string	Spatial coordinate unit	`"km"`, `"m"`, `"cm"`, `"mm"`, `"um"`, `"nm"`	`"m"`
`cmap`	string	Matplotlib colormap	–	`"jet"`
`vlim`	tuple or list	Color scale limits $(v_{\min}, v_{\max})$	–	`None`
`title`	string	Plot title	–	`""`
`xlabel`	bool	Display the x-axis label if `True`	–	`True`
`ylabel`	bool	Display the y-axis label if `True`	–	`True`

If `data` is a NumPy array, it must have the same shape as the computed wavefield. The `unit` parameter changes the displayed coordinate scale. It does not change the underlying simulation grid.

5.4 `error_norm_point_source`

The method `error_norm_point_source` calculates the relative L_2 error, expressed as a percentage, between the numerical point-source solution and the analytical reference field.

```
ho = ph.Helmholtz()
ho.solve()

error_percent = ho.error_norm_point_source()
print(error_percent)
```

This example gives a relative L_2 error of 16.3% for the default configuration: the EM2 absorbing boundary method, second-order finite differences, and grid spacing $h = 0.01$ m (approximately 15 grid points per wavelength).

References

[1] Collino F, Tsogka C. Application of the perfectly matched absorbing layer model to the linear elastodynamic problem in anisotropic heterogeneous media. Geophysics. 2001;66(1):294-307.